NAME: DIVYA.S

REGNO: 22MID0355

TASK-05:

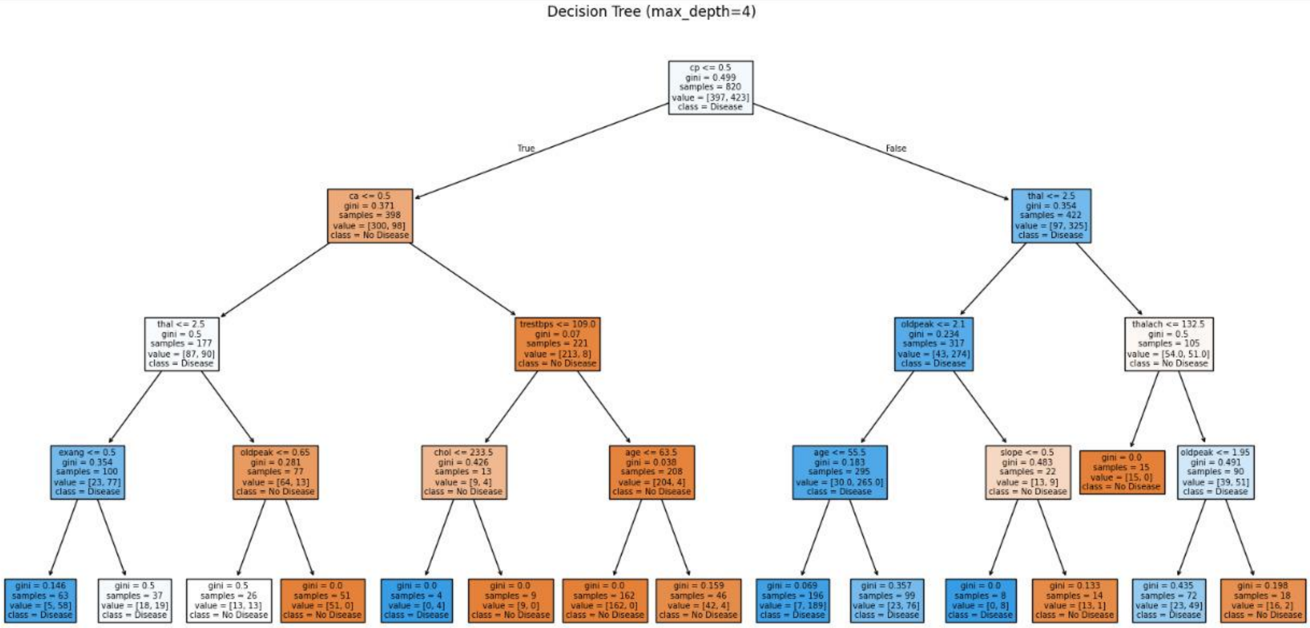
CODE:

```
[7]:
import pandas as pd
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv(r"C:\Users\julug\Downloads\heart disease.csv")
print("First 5 rows:\n", df.head())

print("\nColumns:", df.columns.tolist())

if 'target' not in df.columns:
    raise ValueError("Please replace 'target' with the correct column name.")

df = df.dropna()

X = df.drop(columns=["target"])
y = df["target"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

dt = DecisionTreeClassifier(max_depth=4, random_state=42)
dt.fit(X_train, y_train)

plt.figure(figsize=(20,10))
plot_tree(dt, feature_names=X.columns, class_names=["No Disease", "Disease"], filled=True)
plt.title("Decision Tree (max_depth=4)")
plt.show()

train_acc = dt.score(X_train, y_train)
test_acc = dt.score(X_test, y_test)
print(f"\nDecision Tree - Train Accuracy: {train_acc:.2f}")
print(f"Decision Tree - Test Accuracy: {test_acc:.2f}")
```

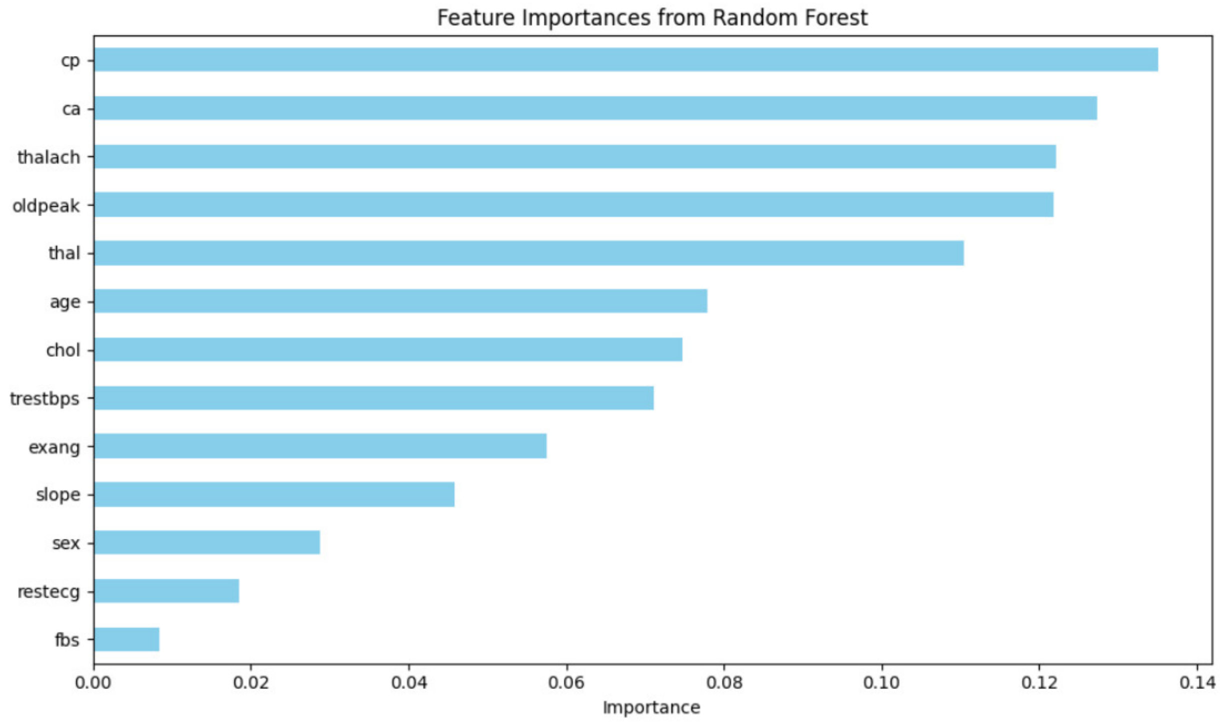
```
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
rf_pred = rf.predict(X_test)

print("\nRandom Forest - Test Accuracy:", accuracy_score(y_test, rf_pred))
print("\nClassification Report (Random Forest):\n", classification_report(y_test, rf_pred))


importances = pd.Series(rf.feature_importances_, index=X.columns)
importances.sort_values().plot(kind='barh', figsize=(10,6), color='skyblue')
plt.title("Feature Importances from Random Forest")
plt.xlabel("Importance")
plt.tight_layout()
plt.show()


dt_cv = cross_val_score(dt, X, y, cv=5)
rf_cv = cross_val_score(rf, X, y, cv=5)

print(f"\nDecision Tree CV Accuracy: {dt_cv.mean():.2f}")
print(f"Random Forest CV Accuracy: {rf_cv.mean():.2f}")
```

OUTPUT:

```
First 5 rows:
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  slope  \
0   52    1   0       125   212    0        1      168      0      1.0      2
1   53    1   0       140   203    1        0      155      1      3.1      0
2   70    1   0       145   174    0        1      125      1      2.6      0
3   61    1   0       148   203    0        1      161      0      0.0      2
4   62    0   0       138   294    1        1      106      0      1.9      1

   ca  thal  target
0   2     3       0
1   0     3       0
2   0     3       0
3   1     3       0
4   3     2       0

Columns: ['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target']
```




```
Decision Tree - Train Accuracy: 0.88
Decision Tree - Test Accuracy: 0.80

Random Forest - Test Accuracy: 0.9853658536585366

Classification Report (Random Forest):
              precision    recall  f1-score   support

           0       0.97      1.00      0.99       102
           1       1.00      0.97      0.99       103

    accuracy                           0.99       205
   macro avg       0.99      0.99      0.99       205
weighted avg       0.99      0.99      0.99       205
```




```
Decision Tree CV Accuracy: 0.83
Random Forest CV Accuracy: 1.00
```